

个人信息

姓名: 吴杰

出生日期: 2002.9.9

电话: 18679205922

本科:

山西大学-物理电子工程学院-电子信息科学与技术

专业排名: 1/60

硕士:

南开大学-电子信息与光学工程学院-电子科学与技术
(推免硕士生)

导师: 徐文涛 (国家杰青)

GPA: (3.56/4)



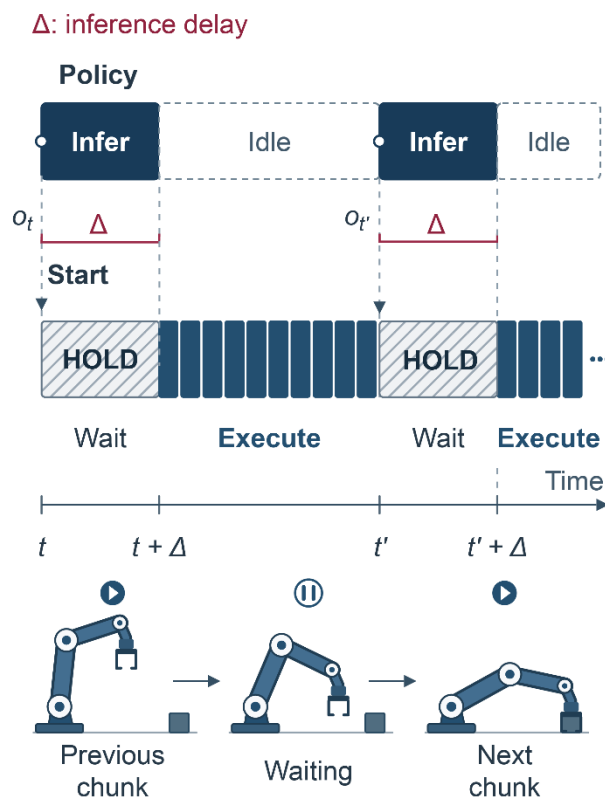
RSA-纠正过时动作

BackGround and Motivation

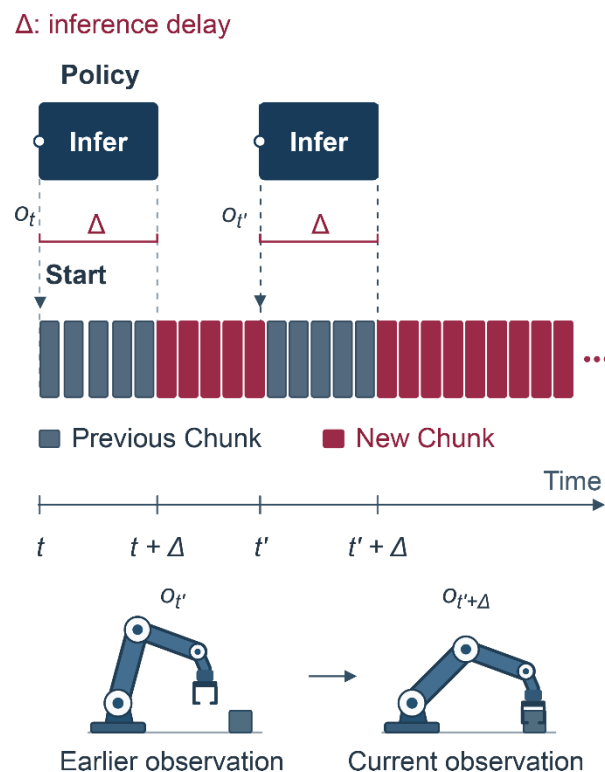
Rectify Stale Action: Latency-Aware Async Inference for Vision-Language-Action

(2027 ICRA 第一作者在投)

(a) Synchronous Inference

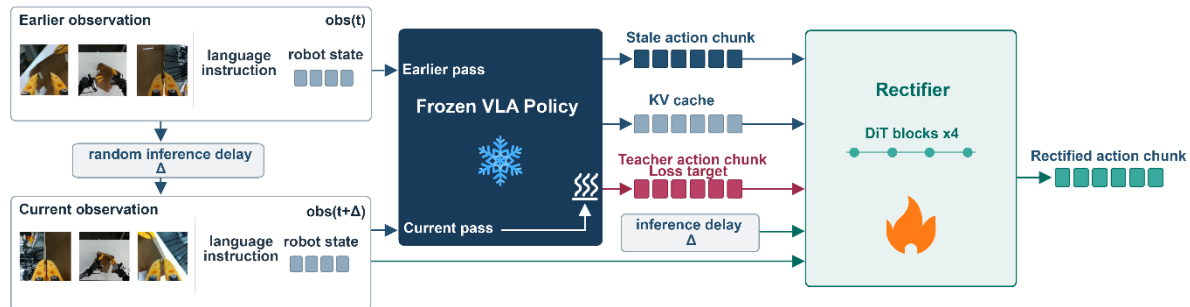


(b) Naive Asynchronous Inference

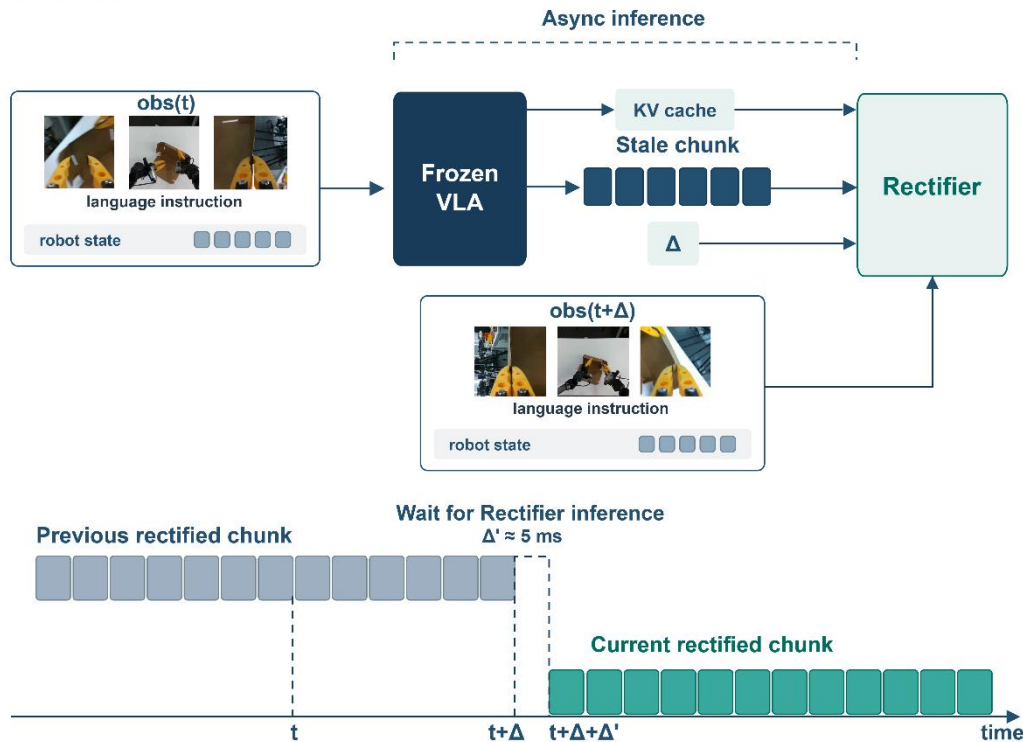


RSA-纠正过时动作 Method

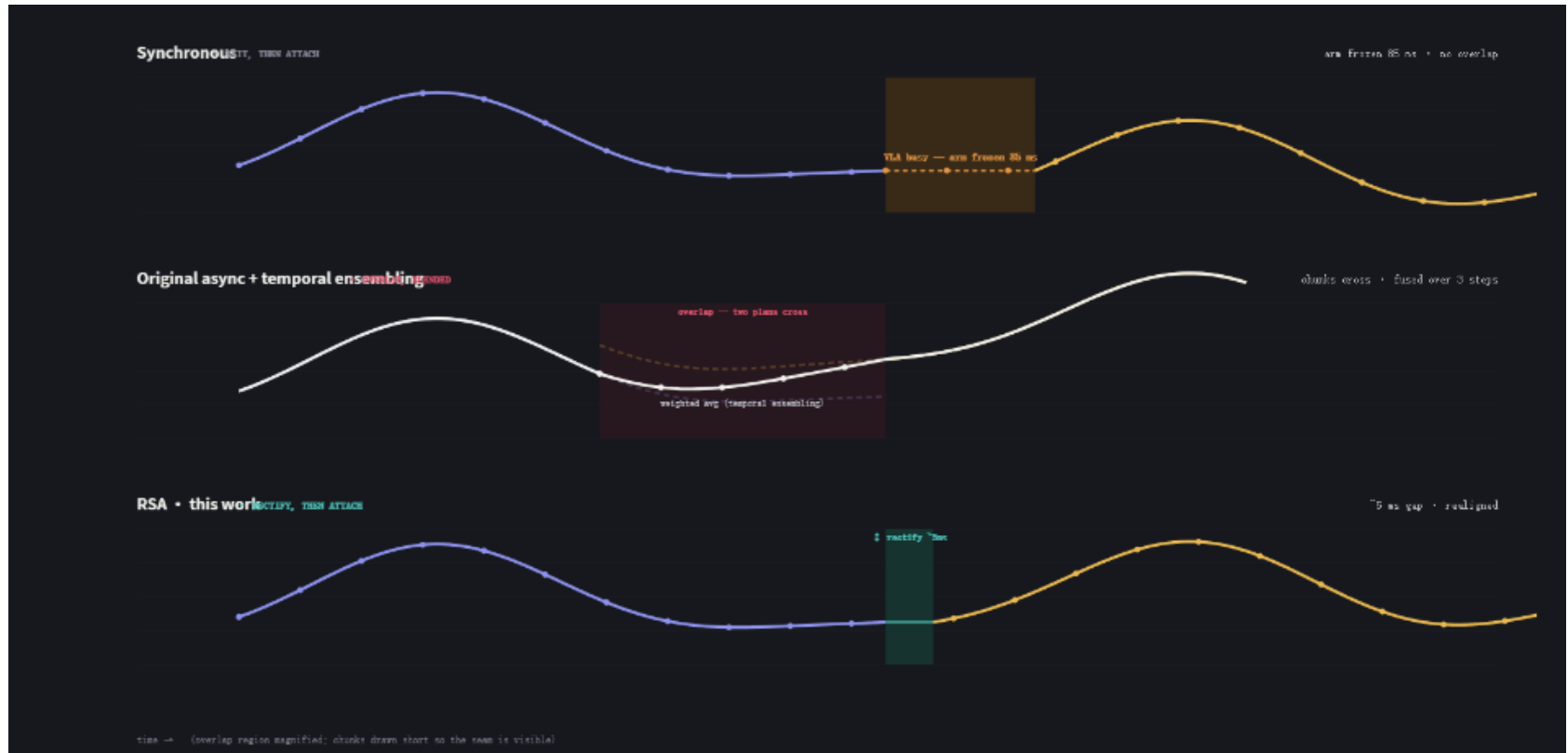
RSA train: VLA-based distillation



RSA inference

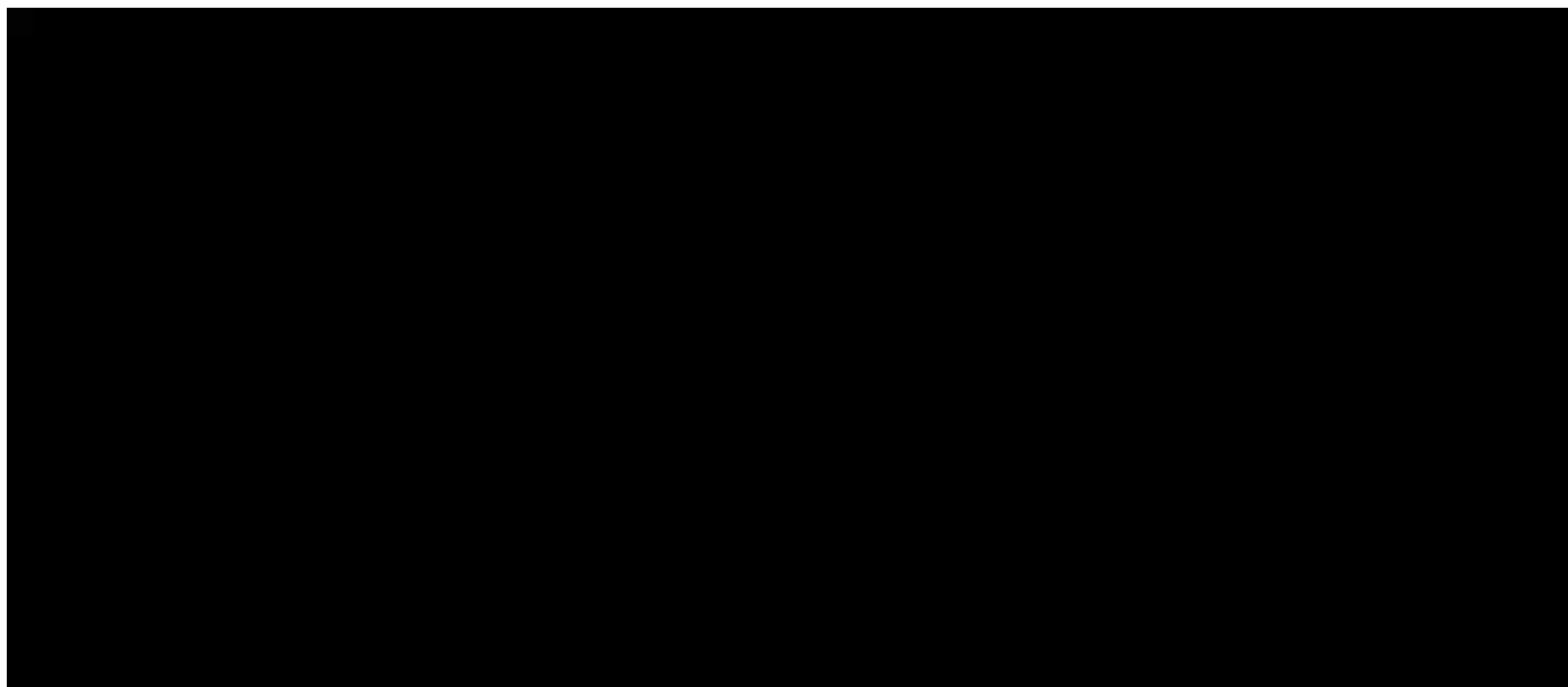


RSA - 纠正过时动作 Method



RSA-纠正过时动作

Video



Forcepi

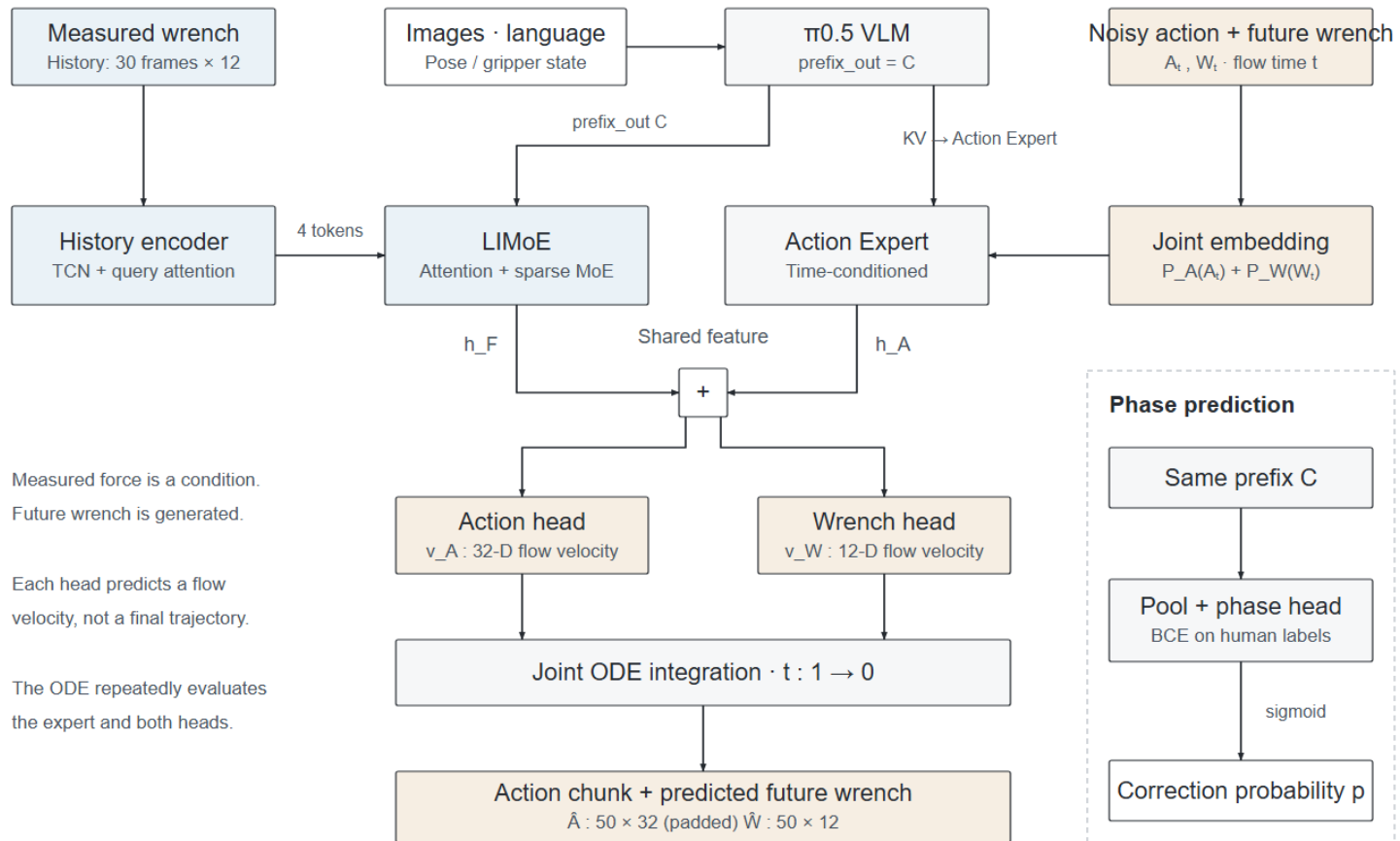


➤基于Gello的3d打印摇操设计

Forcepi

(a) Force0.5-fpred

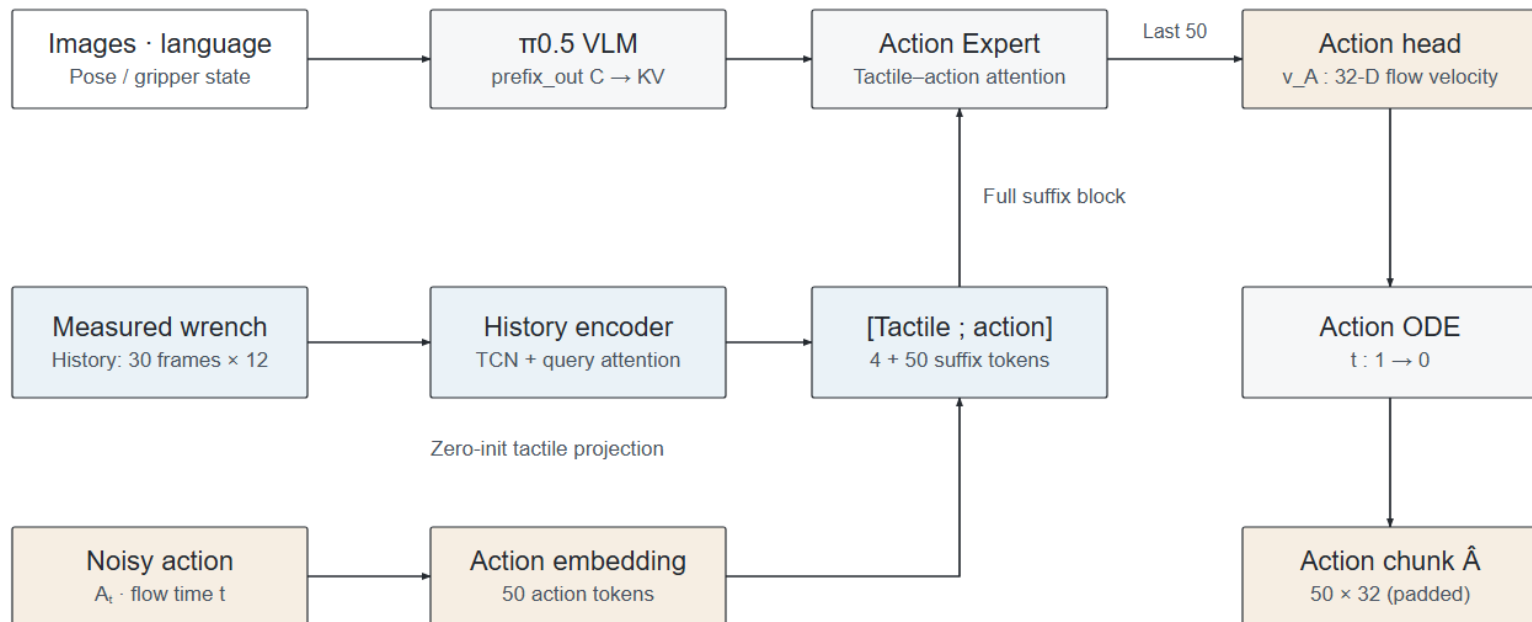
Post-transformer force fusion with joint action-wrench generation



Forcepi

(b) Tac0.5

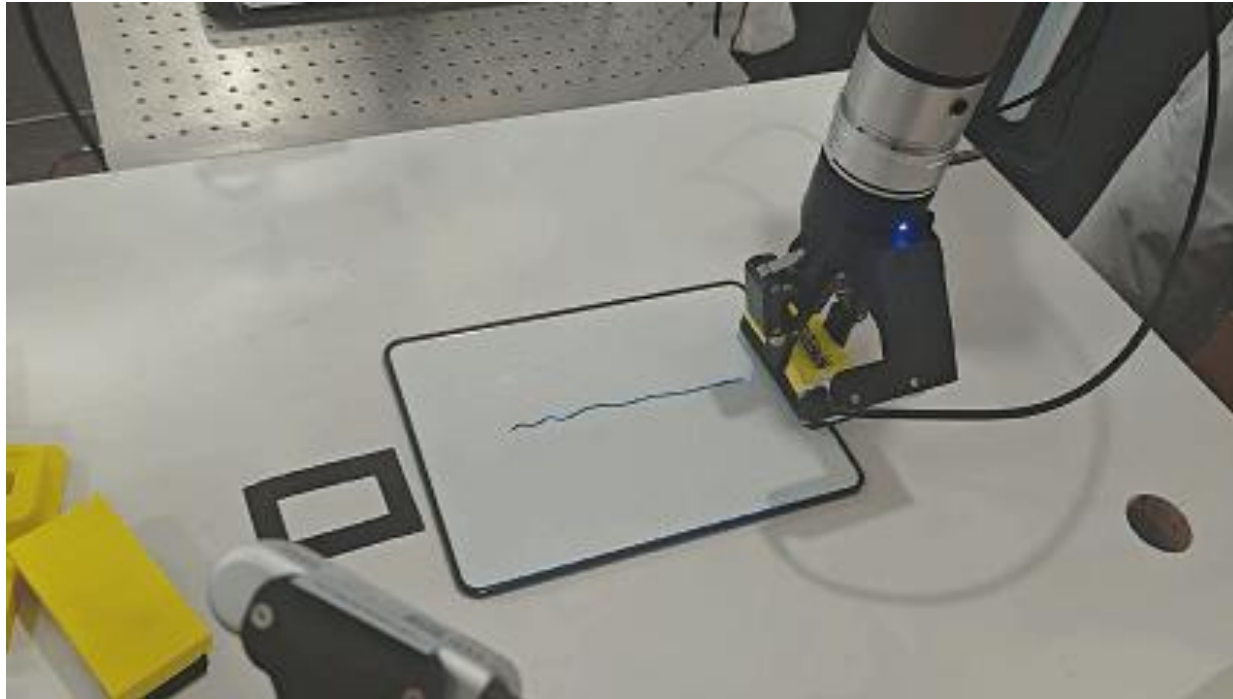
Measured-force tokens enter the Action Expert; the current implementation generates actions only



Force0.5 fuses after the Transformer; Tac0.5 fuses within the Action Expert.

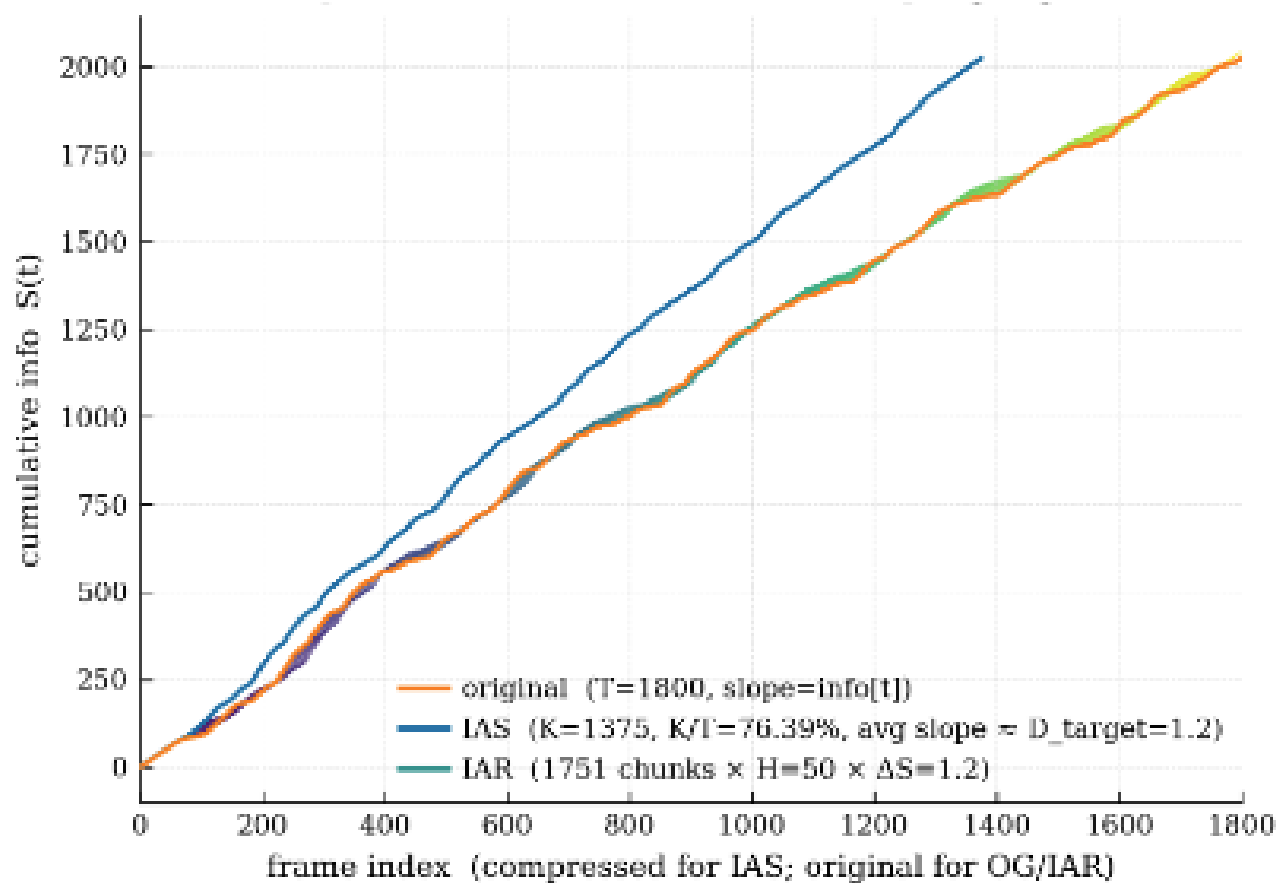
No future-wrench head or phase head is implemented in this Tac0.5 model.

Forcepi



$$M_d(\ddot{x}_d - \ddot{x}_0) + D_d(\dot{x}_d - \dot{x}_0) + K_d(x_d - x_0) = F_{ext}$$

基于Pi0.5全栈式复现 DAR (Direct Action Sampling) 数据处理算法



基于Pi0.5全栈式复现 线性外推控制方法

Top30fps

Left30.1fps

Right30.1fps

Cobot Magic CONNECTED | IP 10.0.3.242 web 8888 roscore 11311 CAN 2

arm launch MIT_V2_vdes.launch 相机序列号 L=346522072780 R=346522075577 F=346522076596

CPU: 15.60% RAM: 34.40/62.60G (55.10%) GPU: 0.00% VRAM: 13.20/48.00G (27.60%)

JOINT MONITOR pos / vel / eff x 8 topics

Oscilloscope

	j0	j1	j2	j3	j4	j5	j6
puppet_C/L	-0.121	1.341	-0.465	-0.090	0.360	-0.128	0.045
vel	0.029	0.011	0.000	0.000	0.028	0.000	0.000
eff	0.000	0.000	0.000	0.000	0.000	0.000	0.110
puppet_T/L	-0.115	1.343	-0.476	-0.096	0.342	-0.119	0.047
vel	0.000	0.000	0.000	0.000	0.000	0.000	0.000
eff	0.000	0.000	0.000	0.000	0.000	0.000	0.000
master_C/L	--	--	--	--	--	--	--
vel	--	--	--	--	--	--	--
eff	--	--	--	--	--	--	--
master_T/L	--	--	--	--	--	--	--
vel	--	--	--	--	--	--	--
eff	--	--	--	--	--	--	--
puppet_C/R	-0.162	1.078	-0.396	0.104	0.685	0.000	0.003
vel	-0.069	-0.057	-0.028	-0.185	0.111	0.032	0.000
eff	0.000	0.000	0.000	0.000	0.000	0.000	-0.881
puppet_T/R	-0.170	1.081	-0.412	0.087	0.672	0.012	0.000
vel	0.000	0.000	0.000	0.000	0.000	0.000	0.000
eff	0.000	0.000	0.000	0.000	0.000	0.000	0.000
master_C/R	--	--	--	--	--	--	--
vel	--	--	--	--	--	--	--
eff	--	--	--	--	--	--	--
master_T/R	--	--	--	--	--	--	--
vel	--	--	--	--	--	--	--
eff	--	--	--	--	--	--	--

END POSE

	x	y	z	qx	qy	qz	qw
Left	0.201	-0.027	0.161	-0.041	0.977	-0.050	0.204
Right	0.146	-0.018	0.181	0.115	0.982	0.026	0.148

INFERENCE COMMAND

SCRIPT my_smoothing

HOST localhost PORT 8000

cd ~/HAOYUE/kai0/
conda activate kai0_inference
python train_deploy_alignment/inference/agilex/inference/agilex_inference_openpi_my_smoothing.py --host localhost --port 8000

CONTROL PANEL

PUPPET JOINT

L j0 0.01 + - Sync Zero

R j0 0.01 + - Sync Zero

MASTER JOINT

L j0 0.01 + - Sync Zero

R j0 0.01 + - Sync Zero

QUICK ACTIONS

一键遥操 (Teleoperate)

一键控制 (Inference)

SYSTEM CONTROL

Puppet En. L ON OFF

Puppet En. R ON OFF

Master En. L ON OFF

Master En. R ON OFF

Config L Master Slave

Config R Master Slave

Teach L ON OFF

Teach R ON OFF

LAUNCH MANAGER

ROS Core RUNNING

RealSense Cameras RUNNING Start Stop

MIT Arm Driver RUNNING Start Stop

File ~/home/ubuntu/anaconda/envs/aloha/lib/python3.8/site-packages/pyAgxArm/protocols/can_protocol/comm/can_comm.py, line 231, in check_can raise self.last_error ValueError: Device 'can_left_mas' does not exist.

CAN ACTIVATE CONFIG activate_can_arms.sh

sudo password for CAN scripts

Find CAN Port STOPPED Start

[sudo] password for ubuntu: ethtool and can-utils are installed.
Interface can_l_slave is on USB port 1-13:1.0
Interface can_r_slave is on USB port 1-6:1.0

CAN Activate STOPPED Start

using usb address: 1-13:1.0
Found interface can_l_slave for USB 1-13:1.0
Interface can_l_slave is already up with bitrate 1000000
Interface name is already can_l_slave
Interface can_l_slave txqueuelen set to 100
-----OVER-----
[sudo] password for ubuntu: -----
-----START-----
ethtool and can-utils are installed.
Using USB address: 1-6:1.0

➤Web_console主控界面

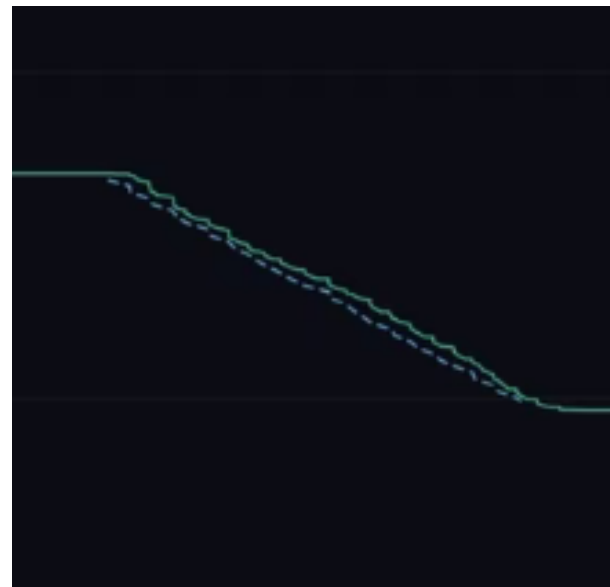
基于Pi0.5全栈式复现 线性外推控制方法



➤ naïve的MIT控制



➤ 加了前馈速度的MIT控制



➤ 加了线性外推的MIT控制

基于Pi0.5全栈式复现

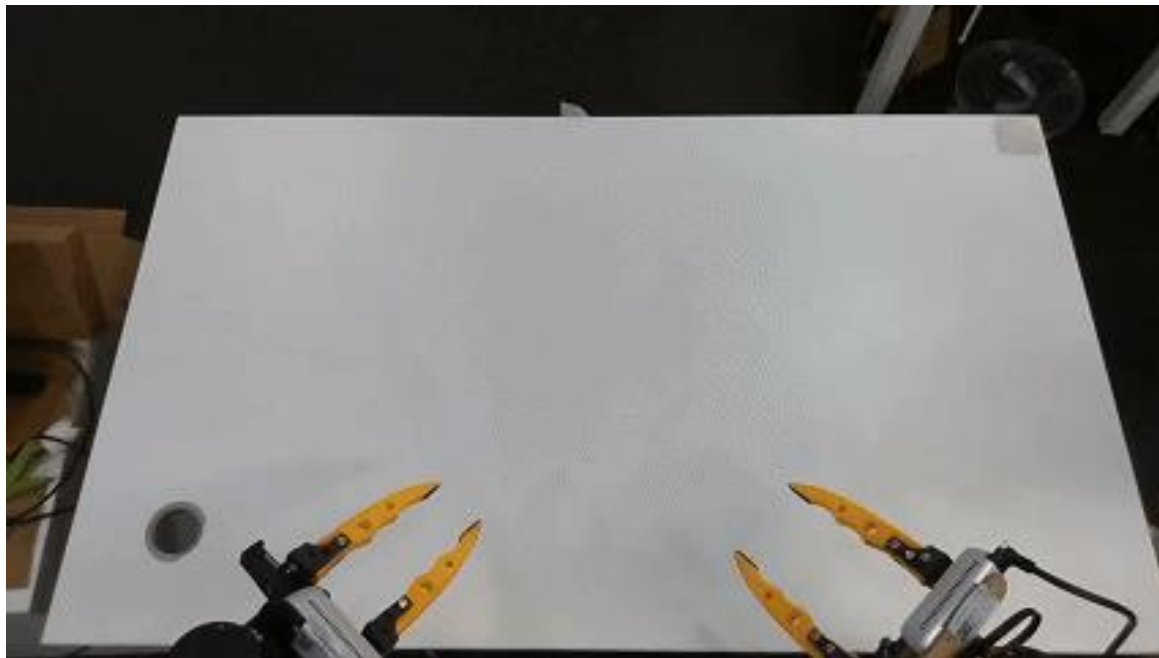


➤ 叠衣服



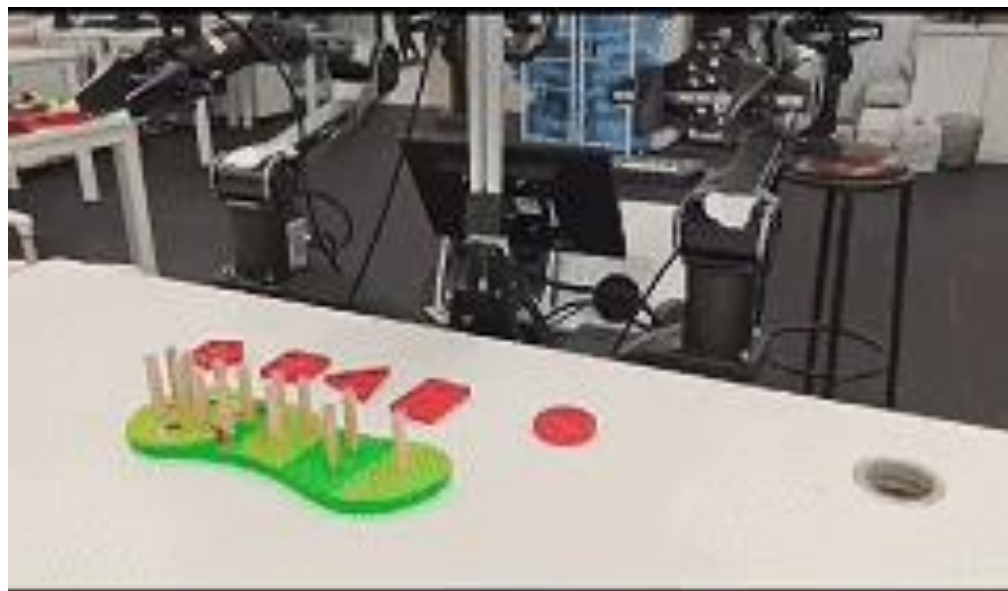
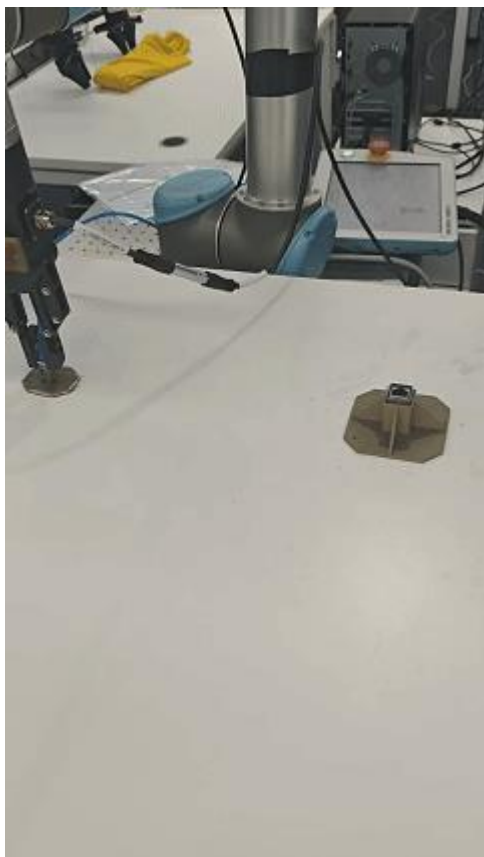
➤ 拔插座

基于Pi0.5全栈式复现 trainingRTC

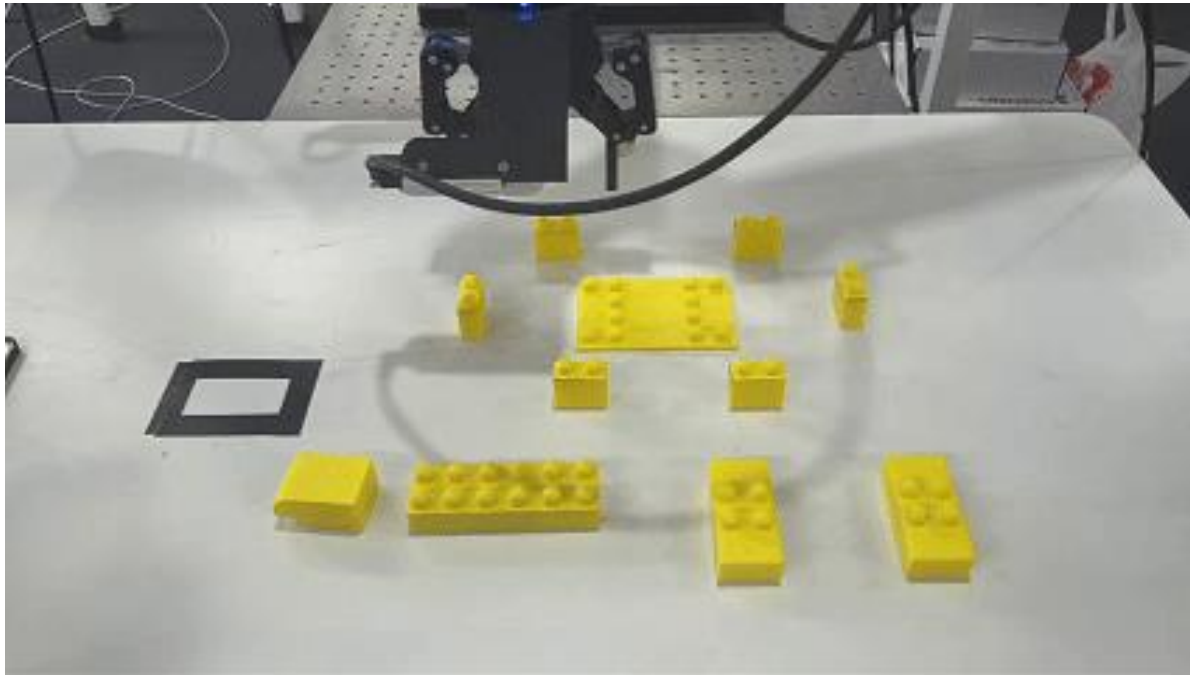


➤ 基于TrainingRTC实现的叠盒子

基于Pi0.5全栈式复现 (真机强化)

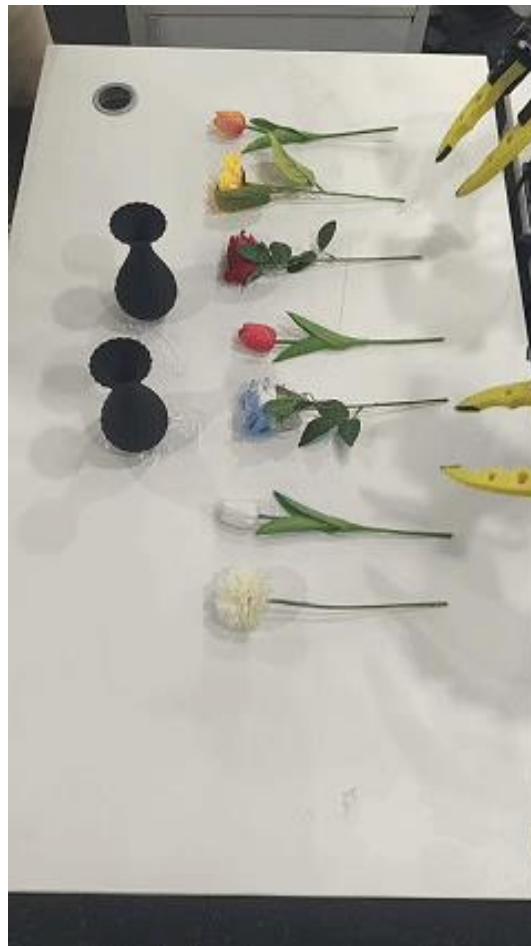


基于Pi0.5全栈式复现 (subtask)



➤ 基于subtask实现的搭积木

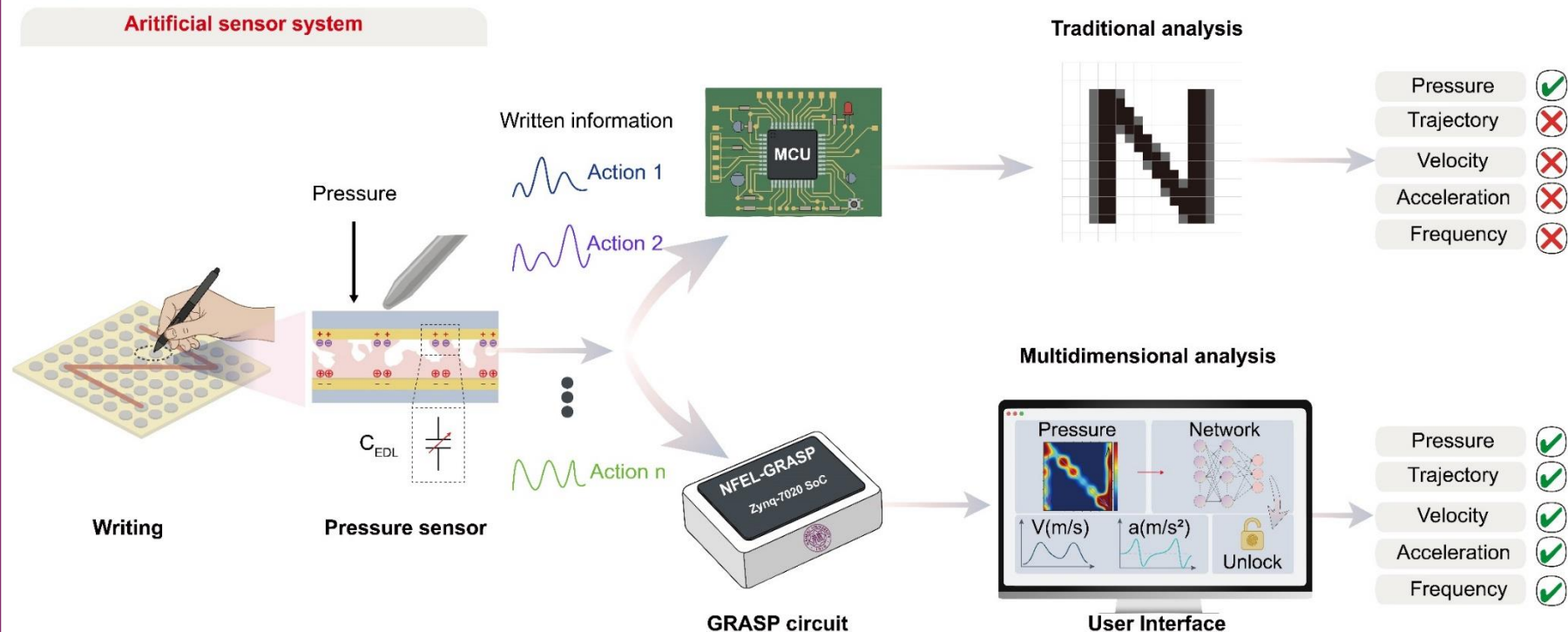
基于Pi0.5全栈式复现 (NativeMEM实现的插花)



➤ 基于NativeMEM实现的人类示教记忆插花

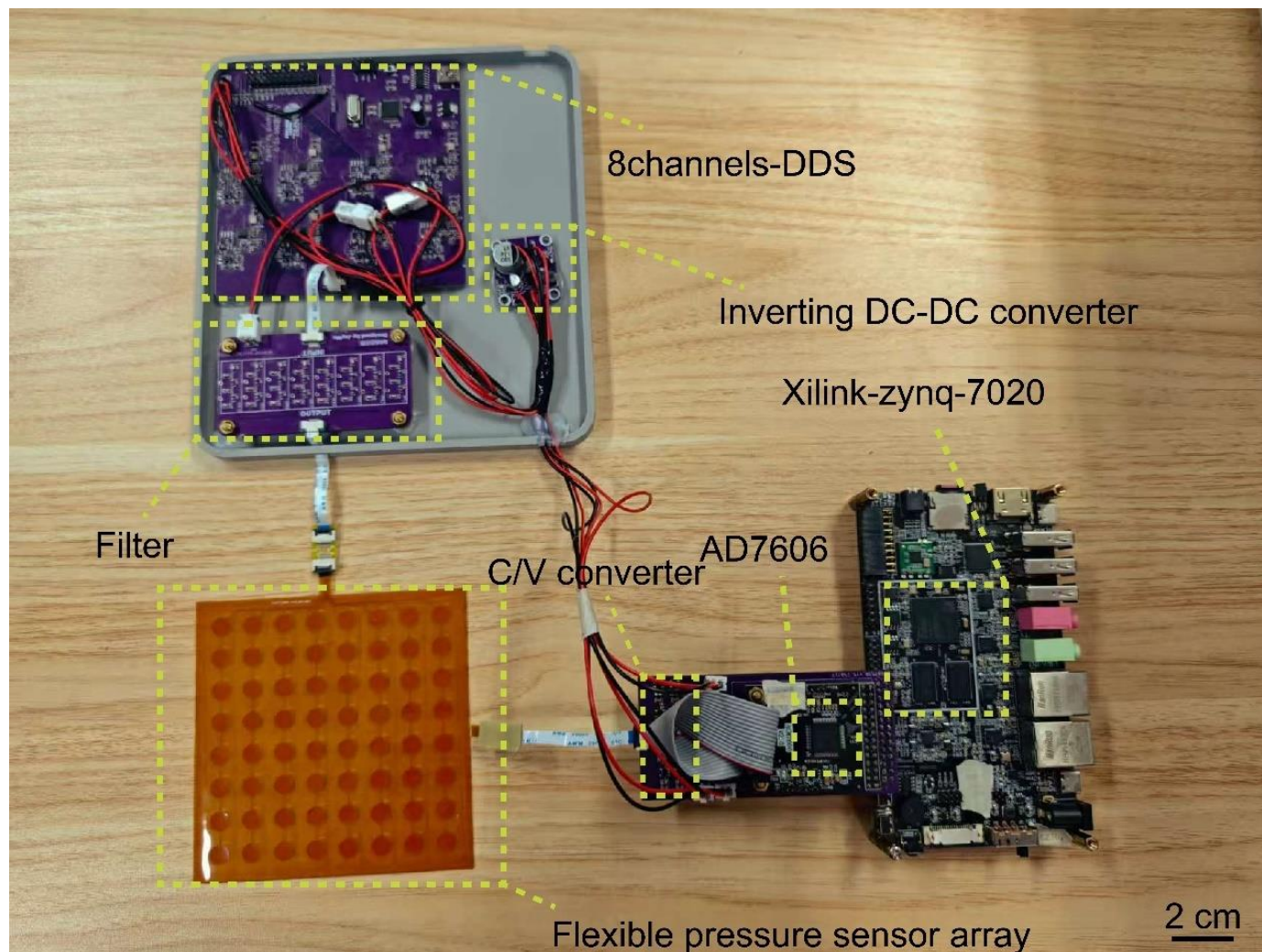
基于FDM架构的手写触觉数据采集系统

➤ GRASP系统感知与传统采集系统对比



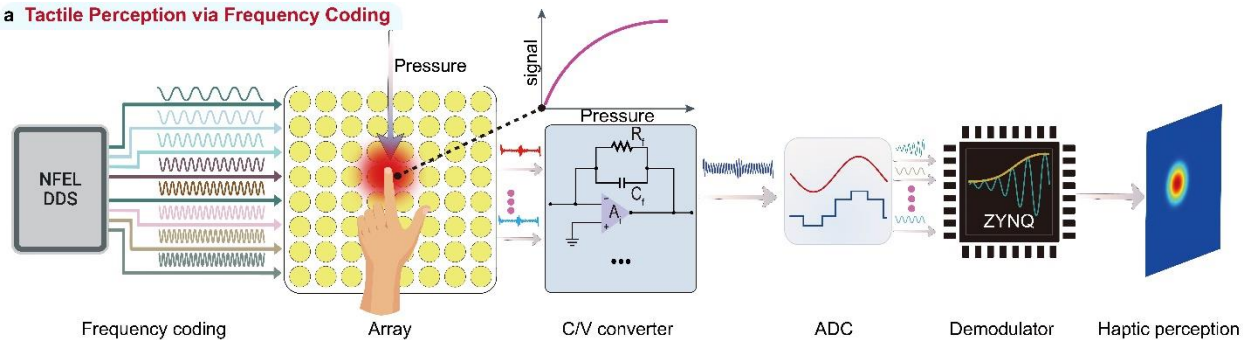
因此研究**行为动态生物识别**具有重要研究意义
此GRASP系统可以测算，**压力，轨迹方向，速度，加速度，频率多维特征。**

基于FDM架构的手写触觉数据采集系统

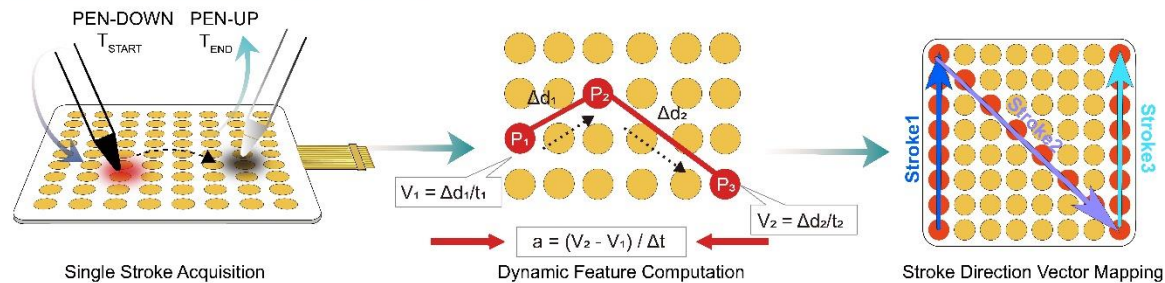


基于FDM架构的手写触觉数据采集系统

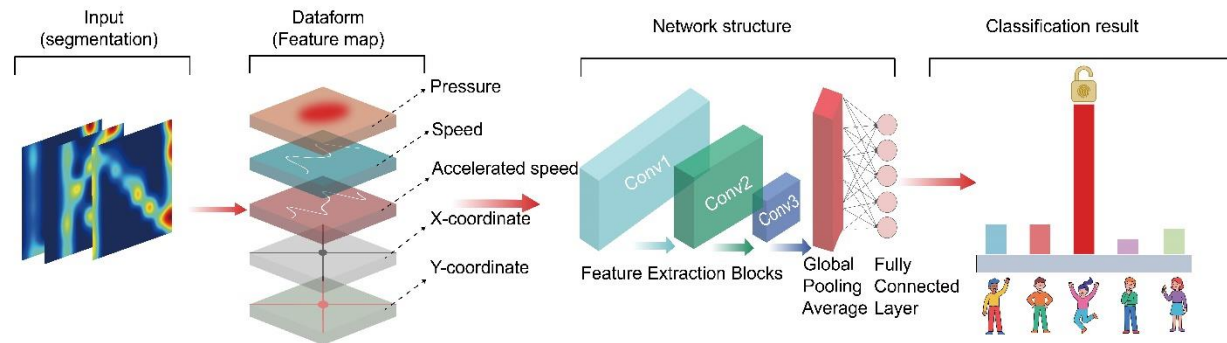
a Tactile Perception via Frequency Coding



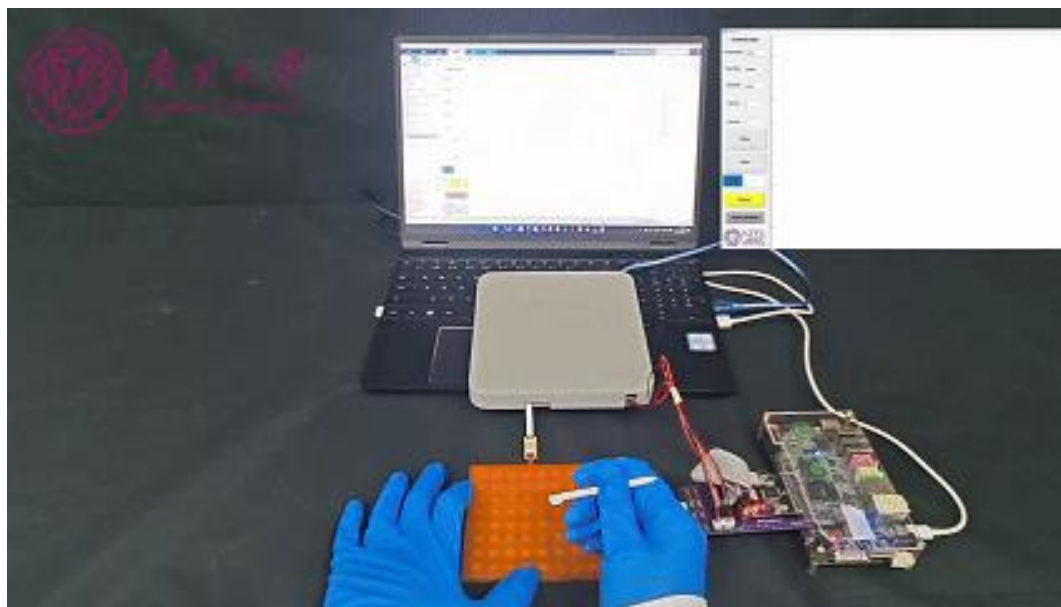
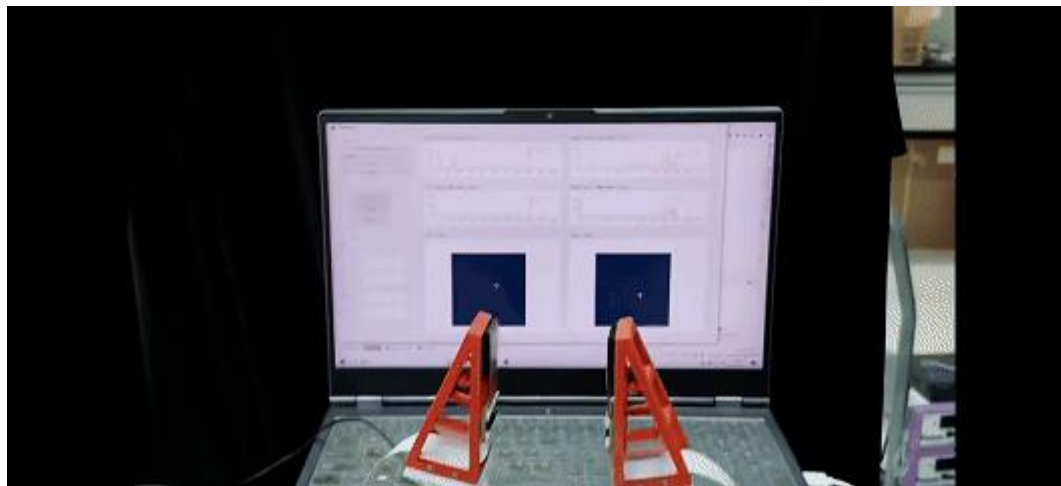
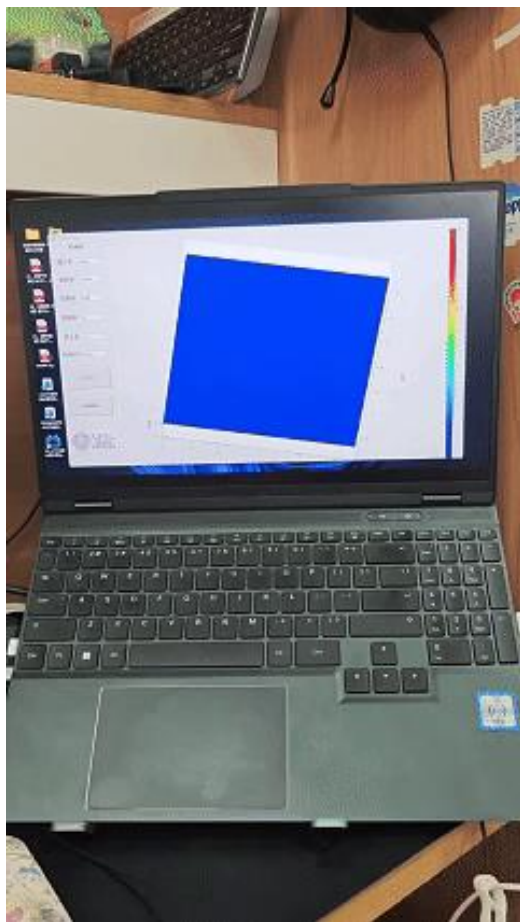
b Multidimensional Handwriting Feature Extraction



c Multi-Feature Fusion Deep Learning Framework



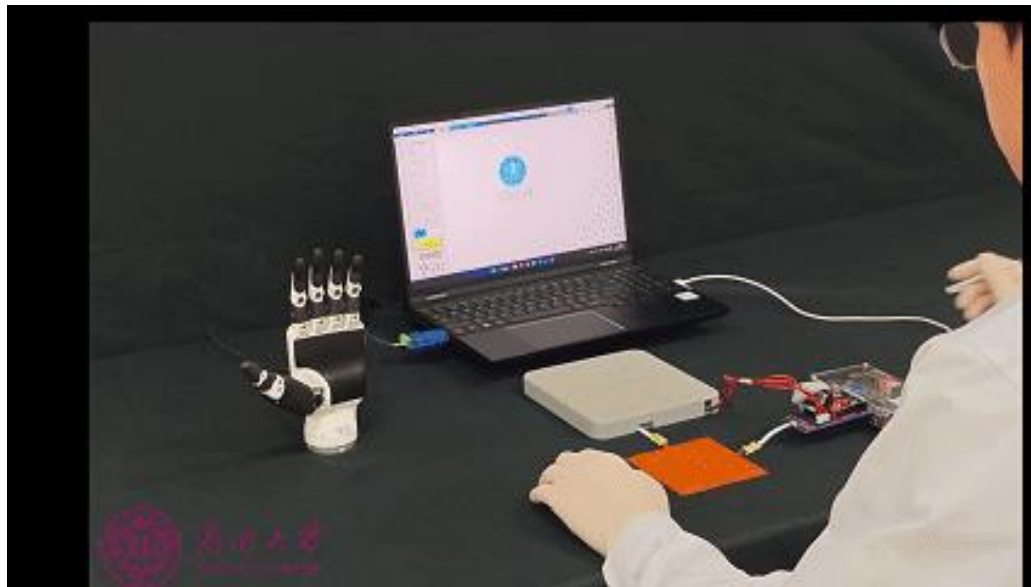
基于FDM架构的手写触觉数据采集系统



基于FDM架构的手写触觉数据采集系统



➤ 3d打印灵巧手装配电子皮肤



➤ 动态生物识别

Dear Mr Wu,

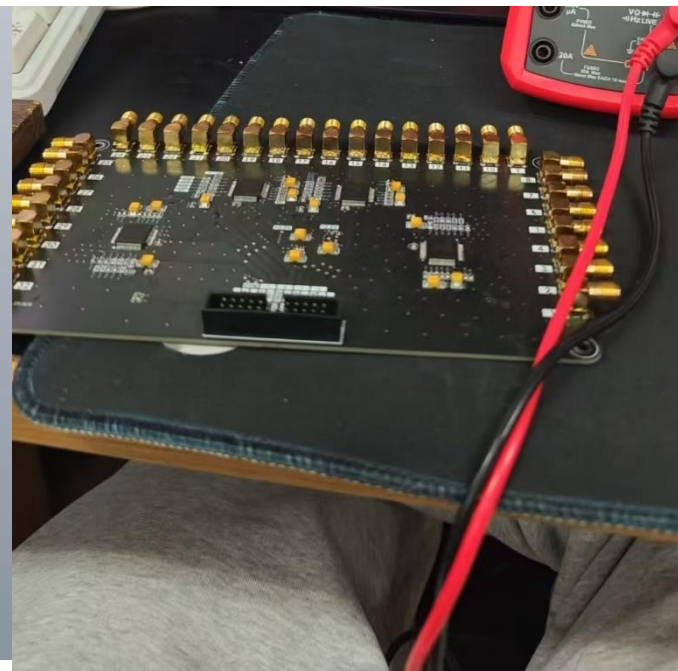
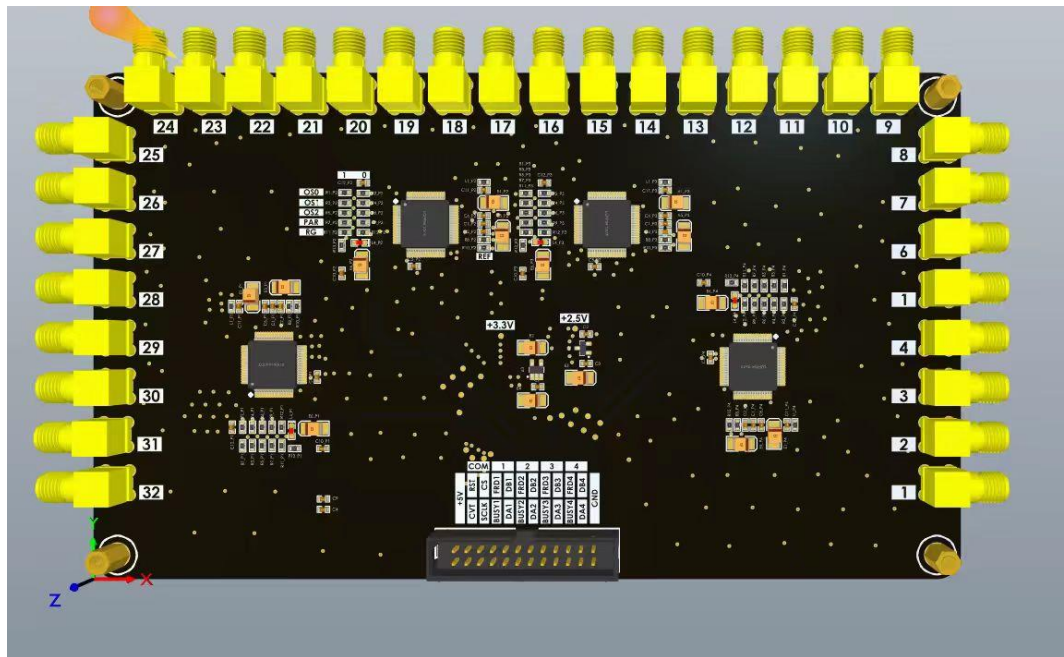
Please note that you are listed as a co-author on the manuscript "Five-Dimensional Dynamically Encrypting Flexible Electronic Skin" (reference number: NATSENSORS-26071437), which was recently submitted to Nature Sensors.

The corresponding author is solely responsible for communicating with the journal and managing communication between co-authors. Please contact the corresponding author directly with any queries you may have related to this



➤ Nature Sensors在投
(第一作者)

基于上述FDM电子皮肤采集做的拓展硬件 32 Channels ADC (AD7606)



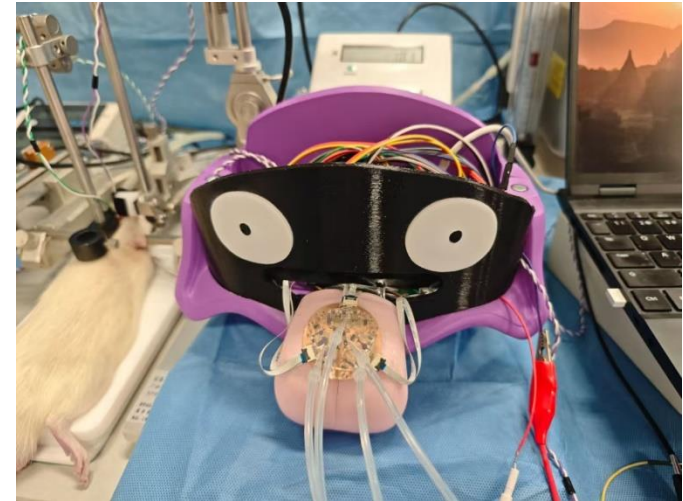
目前已编写对应的stm32f103C8T6驱动，以及相应的FPGA驱动。

人工重构味觉神经网络

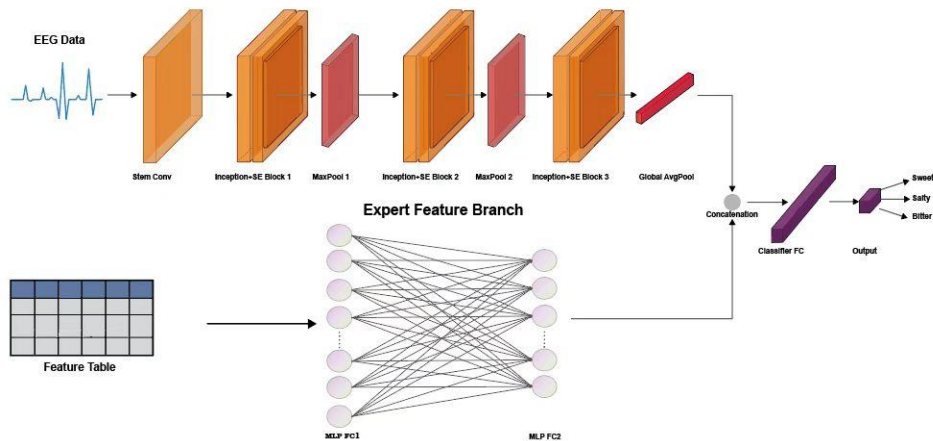
Dear Dr. Wu,

You are listed as a co-author on the manuscript, "An electronic reconstructed gustatory nerve pathway" (2026-05-13189), which was recently submitted to Nature.

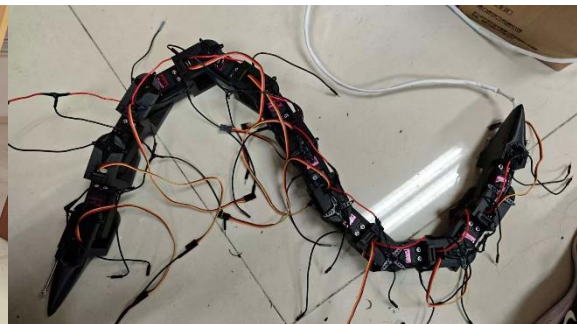
The corresponding author is solely responsible for communicating with the journal and managing communication between co-authors. Please contact the corresponding author directly with any queries you may have related to this manuscript.



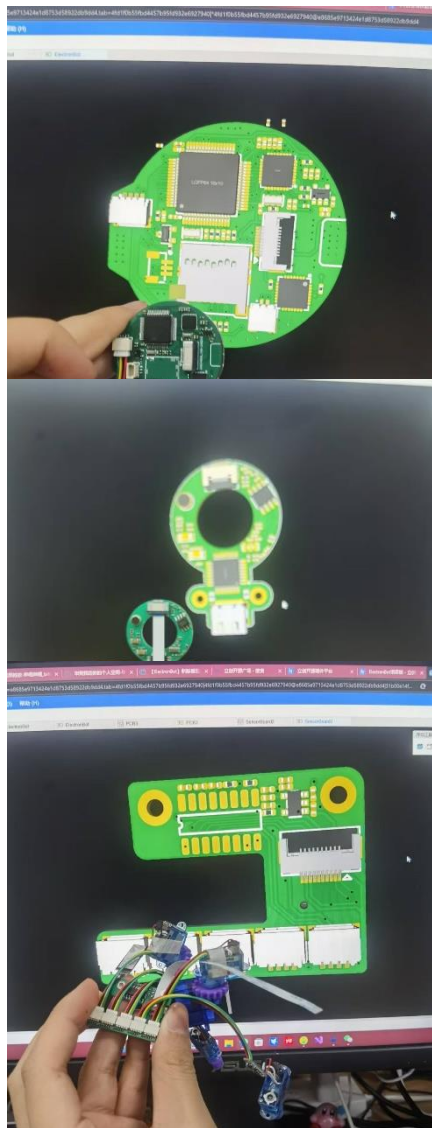
➤ Nature在投(共同第一作者)



额外的个人兴趣



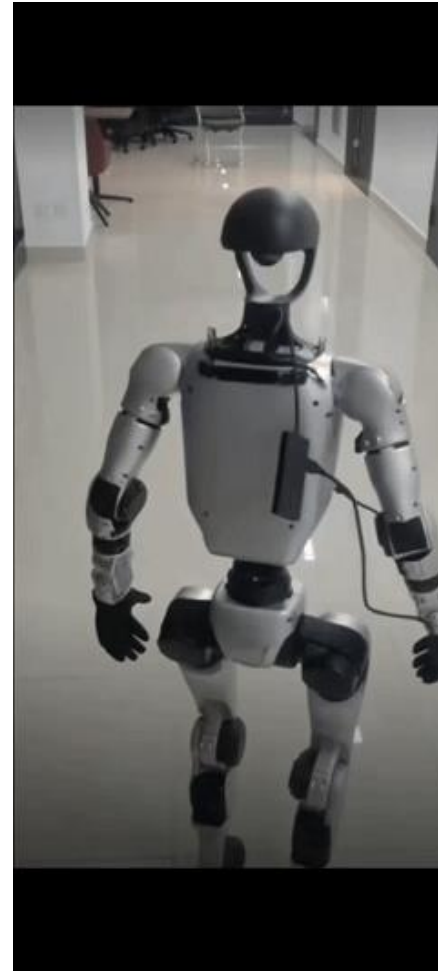
额外的个人兴趣



额外的个人兴趣

➤ 宇树G1 (internVLN复现)

prompt: move forward until you are close to the chair, then turn right to face the door and enter the room. Then stop when you are close to the table.



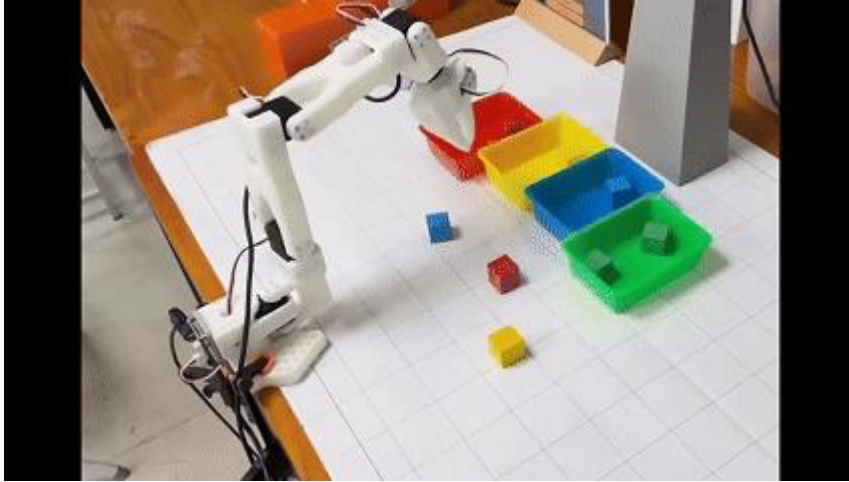
额外的个人兴趣

语音指令：小雨小雨，帮我拿绿色的沙包



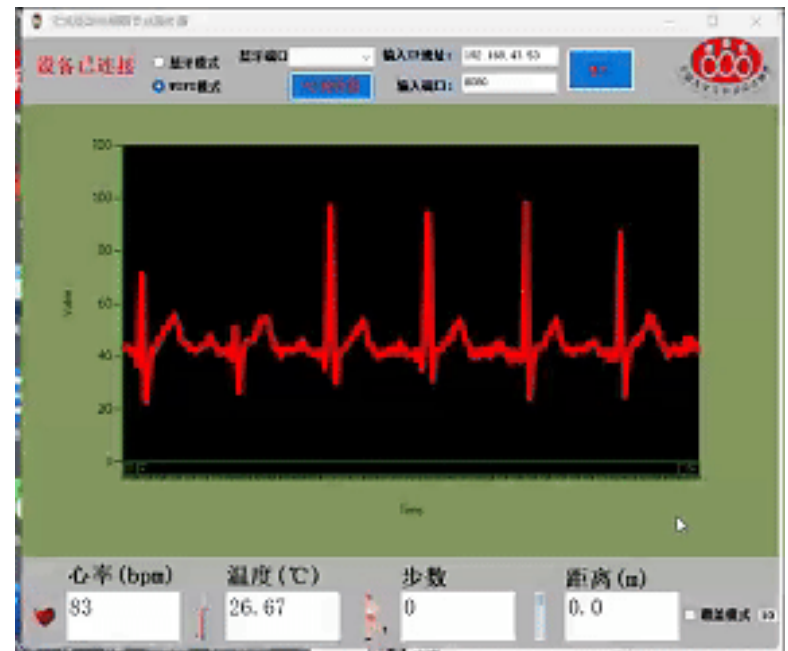
- 首先手眼标定->得到手眼相对矩阵->然后通过VLM对读取图像做分割标注->深度相机计算得到相应物体的坐标（利用点云计算得到质心的位置，就是该物体的最终坐标）->然后IK解算去抓取物体
(语音控制基于科大讯飞API)

额外个人兴趣

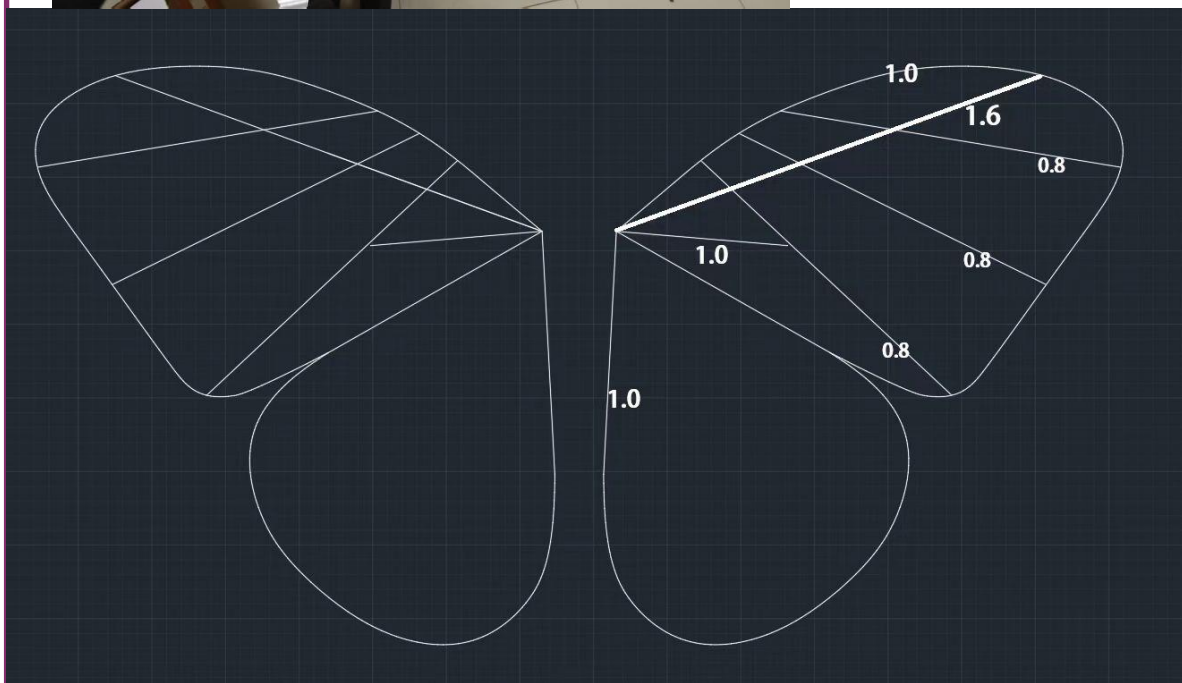


➤ SO101全栈复现

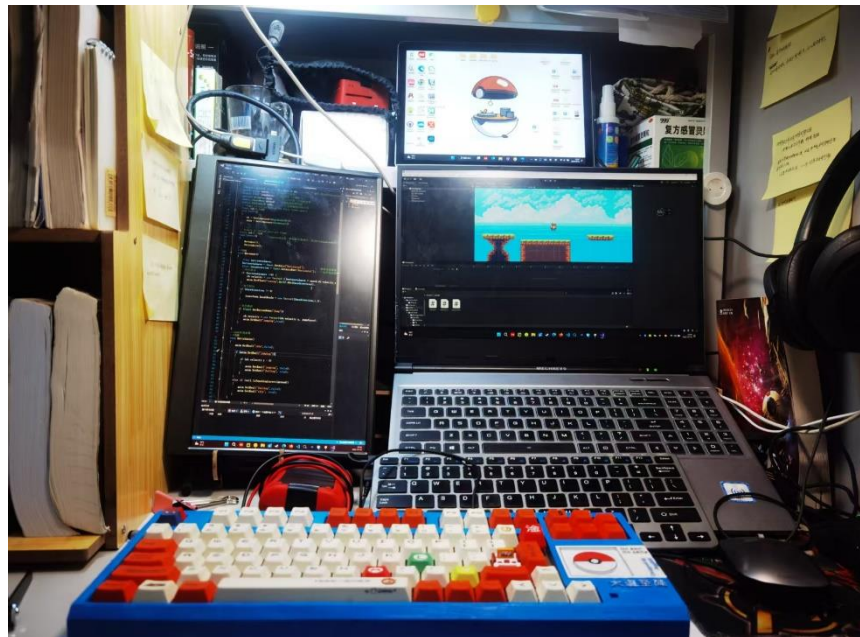
➤ 心电实时监测



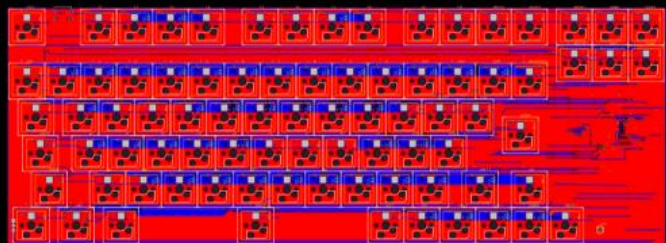
仿生蝴蝶



开源键盘



目前开源获得64个star



开源键盘

